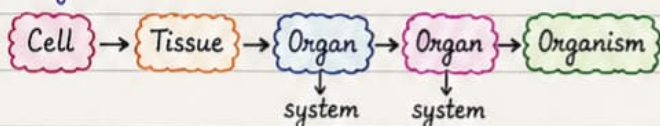


STRUCTURAL Organisation in Animals

- All animals are made up of cells. These cells show a remarkable organisation and arrangement into various tissues, organs and organ systems to perform complex functions.

1. LEVELS OF ORGANISATION

Animals exhibit different levels of organisation -



- Cell** → Structural & functional unit of life.
- Tissue** → Group of similar cells performing a common function.
- Organ** → Made of different tissues working together.
- Organ system** → Group of organs coordinating a major function.
- Organism** → Complete living individual.

2. ANIMAL TISSUES

There are four basic types of animal tissues:

- ① Epithelial tissue
- ② Connective tissue
- ③ Muscular tissue
- ④ Nervous tissue

Points to remember

- Tissues are of two types →
- Simple tissue** → One type of cells
- Complex tissue** → More than one type of cells

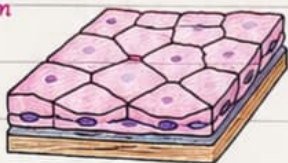
3. EPITHELIAL TISSUE

Epithelial tissue covers the body surface, lines body cavities and forms glands.

Types →

(i) Squamous epithelium

- Single layer of thin, flat cells.
- Function → Diffusion & filtration.
- Example → Alveoli of lungs, lining of blood vessels.



Squamous epithelium

(ii) Cuboidal epithelium

- Cube-shaped cells.
- Function → Secretion & absorption.
- Example → Kidney tubules, ducts of glands.



Cuboidal epithelium

(iii) Columnar epithelium

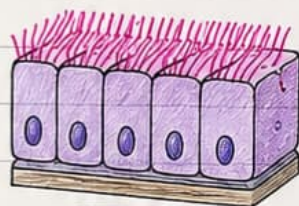
- Tall, column-like cells.
- Function → Absorption & secretion.
- Example → Inner lining of stomach, intestine.



Columnar epithelium

(iv) Ciliated epithelium

- Columnar cells with hair-like cilia.
- Function → Movement of mucus or particles.
- Example → Trachea, fallopian tubes.



Ciliated epithelium

Note Epithelial tissue has little intercellular space and rests on a basement membrane. It is avascular but innervated.

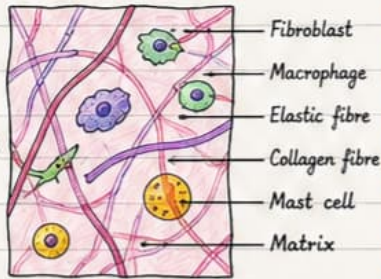
4. CONNECTIVE TISSUE

Connective tissues are the most abundant and widely distributed tissues in animals.

TYPES OF CONNECTIVE TISSUE

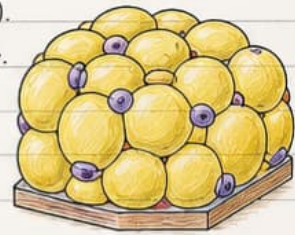
(i) Loose (Areolar) Connective Tissue

- Most widely distributed.
- Cells are loosely arranged in a fluid matrix.
- Found beneath epithelia, around blood vessels and nerves.
- Example → Packing tissue beneath skin.



(ii) Adipose Tissue

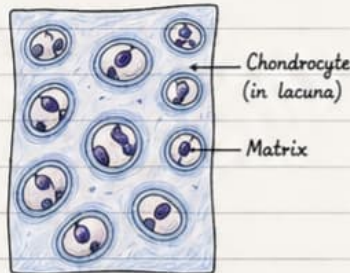
- Cells store fat (adipose cells).
- Acts as an energy reservoir.
- Provides insulation and cushioning.
- Example → Beneath skin, around kidneys, heart.



Adipose tissue (Fat cells)

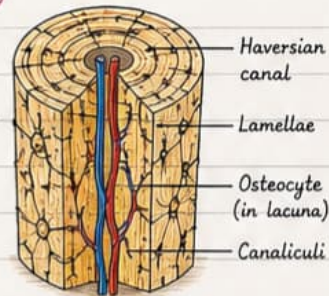
(iii) Cartilage

- Firm but flexible connective tissue.
- Cells called chondrocytes lie in lacunae.
- Matrix is firm due to presence of chondrin.
- Example → Tip of nose, pinna of ear, trachea.



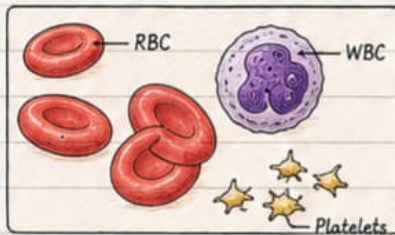
(iv) Bone (Osseous Tissue)

- Hardest and strongest connective tissue.
- Matrix is calcified with calcium salts.
- Cells called osteocytes lie in lacunae.
- Function → Provides support and protection.



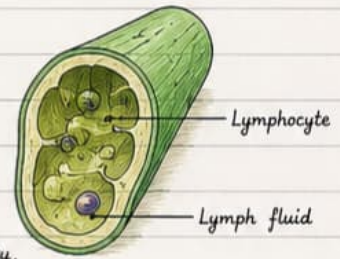
(v) Blood

- Fluid connective tissue.
- Matrix is plasma.
- Cells → RBCs, WBCs, Platelets.
- Function → Transport of gases, nutrients, hormones, wastes and protection.



(vi) Lymph

- Similar to blood but without RBCs.
- Contains mainly lymphocytes.
- Function → Returns tissue fluid to blood and helps in immunity.

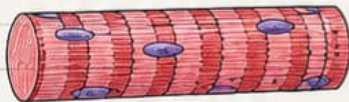


5. MUSCULAR TISSUE

Muscular tissues are specialized for contraction and movement of body parts and internal organs.

(i) Skeletal (Striated) Muscle

- Long, cylindrical, unbranched fibres.
- Striated (striped) appearance.
- Multinucleate with nuclei at periphery.
- Voluntary in action.
- Attached to bones.
- Function → Body movements, maintain posture.



(Striations present)

(ii) Smooth (Unstriated) Muscle

- Spindle-shaped, unbranched fibres.
- No striations.
- Single, central nucleus.
- Involuntary in action.
- Present in walls of internal organs (intestine, stomach, blood vessels, uterus etc.).
- Function → Movement of internal organs.



(No striations)

(iii) Cardiac Muscle

- Branched fibres with striations.
- Single (sometimes two) central nucleus.
- Involuntary in action.
- Found only in heart.
- Function → Rhythmic contraction of heart.

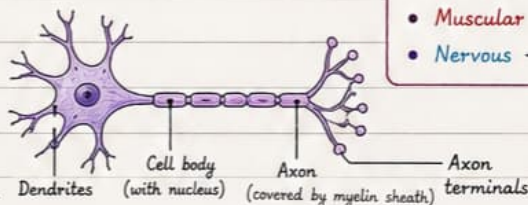


(Striations present)

Note → Muscular tissue is highly vascular and contains contractile proteins (actin & myosin) which bring about contraction.

6. NERVOUS TISSUE

- Made up of neurons (nerve cells) and neuroglia.
- Neurons are specialized for conducting electrical impulses.
- Forms brain, spinal cord and nerves.
- Function → Control and coordination of body activities.



KEY POINTS

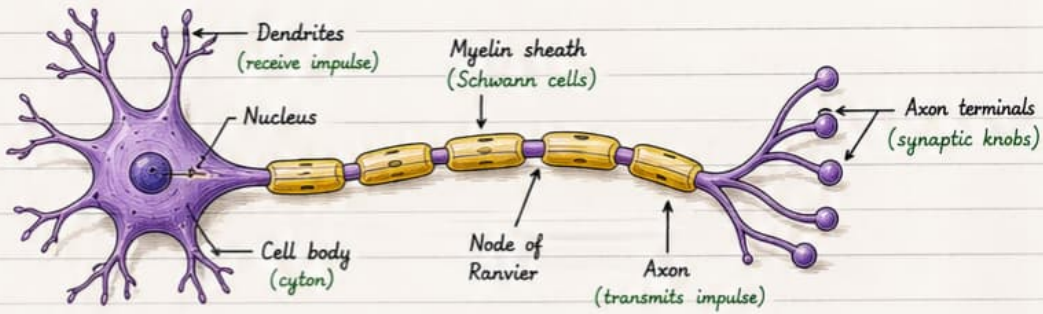
- Epithelial → Covering & glands
- Connective → Binding & support
- Muscular → Movement
- Nervous → Control & coordination

7. NERVOUS TISSUE (contd.)

Nervous tissue has two main cell types → **Neurons** and **Neuroglia** (glial cells).

★ (i) Neuron (Nerve cell)

- Structural & functional unit of nervous tissue.
- Excitable in nature.
- Specialized for conducting nerve impulses.
- Parts of a neuron →



Types of Neurons (Based on function)

1. Sensory (Afferent) Neuron

- Carry impulses from receptors to CNS.
- Cell body lies on the side of the axon.
- Example → Neuron from skin to spinal cord.



2. Motor (Efferent) Neuron

- Carry impulses from CNS to effector organs (muscles / glands).
- Cell body lies at one end of the axon.
- Example → Neuron from spinal cord to muscle.



3. Interneuron (Association) Neuron

- Connects sensory and motor neurons.
- Present entirely within CNS.
- Example → Neurons in spinal cord & brain.



★ (ii) Neuroglia (Glial cells)

- Do not generate or transmit impulses.
- Provide support, protection and nourishment to neurons.
- Types of neuroglia →

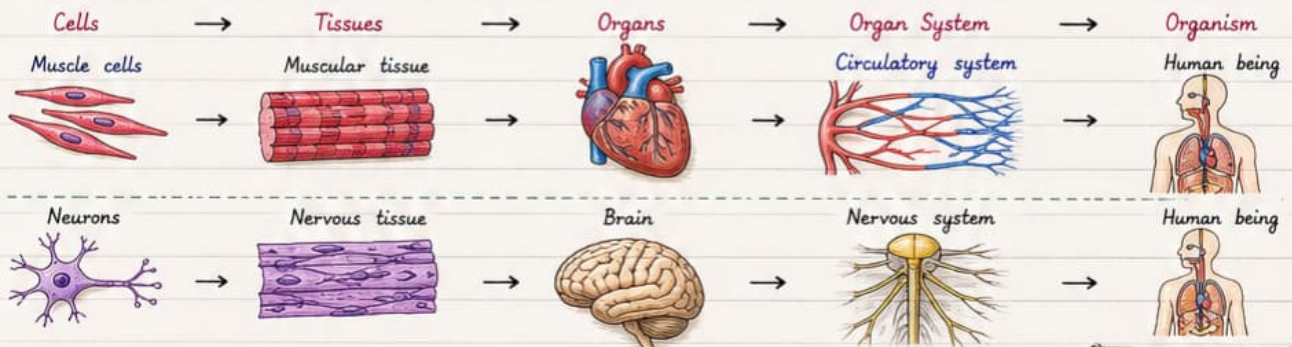
Type	Function
1. Astrocytes	Provide structural support, maintain ionic balance, help in repair.
2. Oligodendrocytes	Form myelin sheath around axons in CNS.
3. Schwann cells	Form myelin sheath around axons in PNS.
4. Microglia	Act as phagocytes; remove cell debris and pathogens.
5. Ependymal cells	Line ventricles of brain and central canal of spinal cord; produce and circulate CSF.

8. ANIMAL ORGANISATION

In multicellular animals, various tissues combine to form organs and organ systems for efficient functioning.

- **Cells** → Basic structural & functional unit of life.
- **Tissues** → Group of similar cells performing a common function.
- **Organs** → Different tissues working together to perform a specific function.
- **Organ Systems** → Group of organs coordinating a major body function.
- **Organism** → Complete living individual. All organ systems work in coordination to maintain life.

★ 9. EXAMPLES OF ORGANISATION IN ANIMALS



Remember!

Structural organisation in animals shows a beautiful hierarchy, from tiny cells to complex systems. Coordination and specialization at each level make life possible.

For NEET

- Focus on tissue types and their functions.
- Diagrams help in better understanding.
- NCERT line based answers are important.

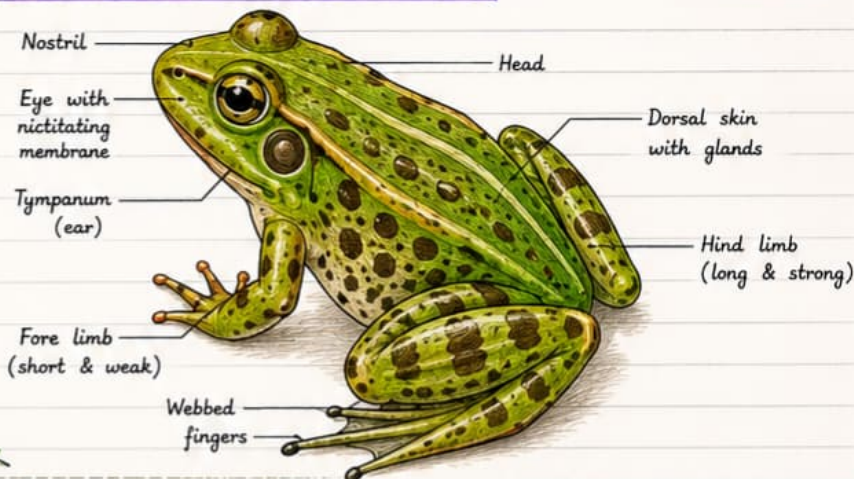
Frog - A Model Organism

Frog (*Rana tigrina*)
a widely studied
amphibian.

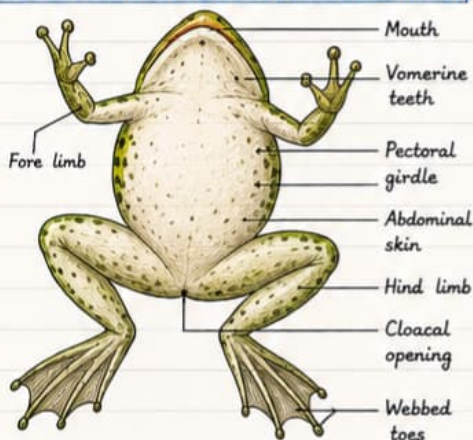
1. GENERAL FEATURES

- Frogs are **amphibians** found in both **aquatic** and **terrestrial** habitats.
- Body is **bilaterally symmetrical** and **dorsoventrally flattened**.
- Skin is **smooth, moist and glandular**; helps in **respiration**.
- Frogs are **ectothermic (cold-blooded)**.
- They show **sexual dimorphism**.
- Fertilization is **external** and development is **indirect**.

2. EXTERNAL FEATURES (DORSAL VIEW)

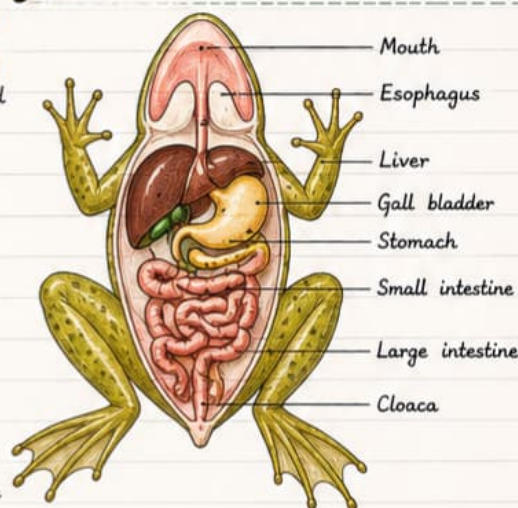


3. EXTERNAL FEATURES (VENTRAL VIEW)



4. DIGESTIVE SYSTEM

- Mouth is wide, provided with **maxillary** and **vomerine** teeth.
- Food is caught by **protrusible, sticky tongue**.
- Alimentary canal** is complete.
- Liver** is large with **gall bladder**.
- Cloaca** is a common chamber for **digestive, urinary and reproductive** systems.



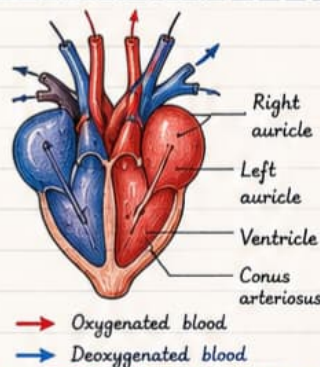
5. RESPIRATORY SYSTEM

- Frogs respire through, **lungs** and **buccopharyngeal cavity**.
- Thin, moist skin** helps in **cutaneous respiration**.
- Lungs** are well developed, **sac-like**.
- During breathing, **floor of mouth** is lowered to draw air in.



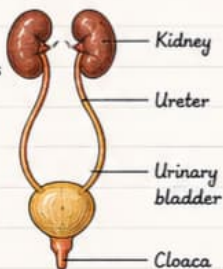
6. CIRCULATORY SYSTEM

- Heart** is **three chambered** (Two auricles & one ventricle).
- Blood** is **mixed** (oxygenated + deoxygenated).
- Double circulation** is **incomplete**.
- RBCs** are **oval** and **nucleated**.



7. EXCRETORY SYSTEM

- A pair of **kidneys** remove **nitrogenous wastes**.
- Urine** passes through **ureters** to **cloaca**.
- Urinary bladder** stores **urine** temporarily.



8. REPRODUCTIVE SYSTEM

Male

- A pair of **testes** lie near **kidneys**.
- Sperm ducts (vasa efferentia)** open into **cloaca**.
- Males** have **vocal sacs** used in **croaking**.



Female

- A pair of **ovaries** lie in the **body cavity**.
- Oviducts** open into **cloaca**.
- Females** lay **eggs** in **water**.



Key Takeaway

Frog is a classic example of **chordate** structural organisation showing the basic plan of body systems in **vertebrates**.

Why Frog is Important?

- ✓ Easy to study
- ✓ Shows fundamental vertebrate features
- ✓ Helps in understanding human biology

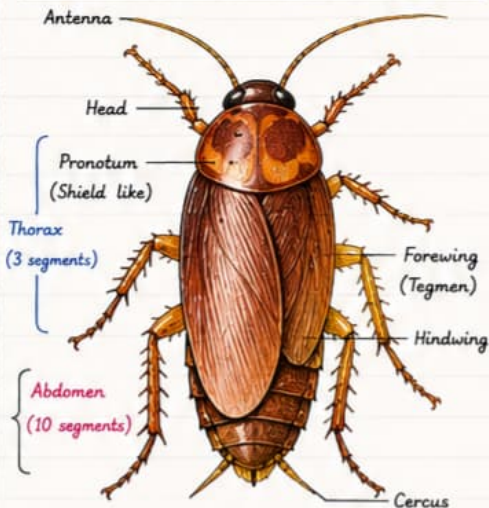
Cockroach - A Model Organism

Cockroach (*Periplaneta americana*) is a commonly studied insect. It shows bilateral symmetry and triploblastic organisation.

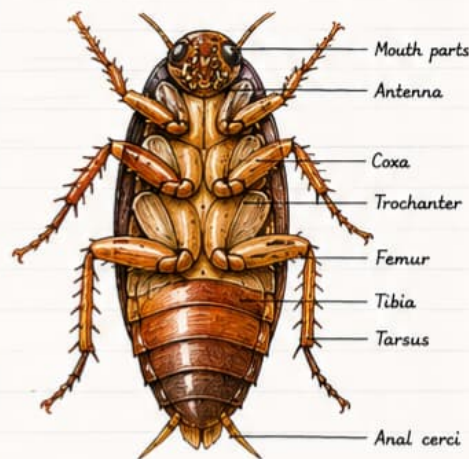
1. GENERAL FEATURES

- Belongs to Phylum *Arthropoda* and Class *Insecta*.
- Body is bilaterally symmetrical and segmented.
- Body divided into head, thorax and abdomen.
- Three pairs of jointed legs present on thorax.
- Two pairs of wings (forewings leathery, hindwings membranous).
- One pair of long, filamentous antennae used for sensing.

2. EXTERNAL MORPHOLOGY (DORSAL VIEW)

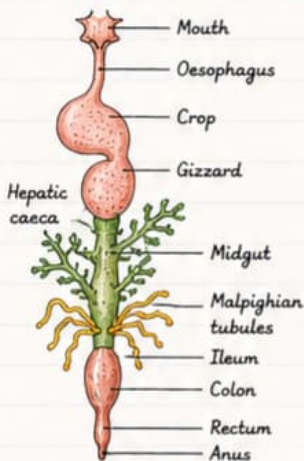


3. EXTERNAL MORPHOLOGY (VENTRAL VIEW)



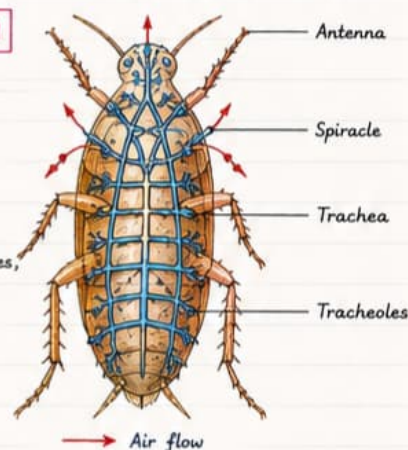
4. DIGESTIVE SYSTEM

- Alimentary canal is complete.
- Mouth parts are of biting and chewing type.
- Food is chewed, mixed with saliva and swallowed.
- Hepatic caeca help in digestion and absorption.
- Undigested food is removed through anus.



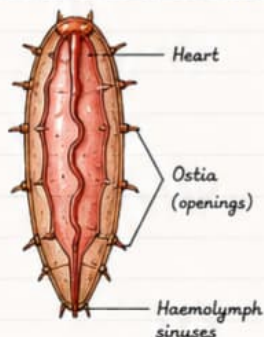
5. RESPIRATORY SYSTEM

- Respiration takes place through tracheal system.
- Spiracles are present on lateral sides of thorax and abdomen.
- Air enters through spiracles, travels in tracheae and tracheoles and reaches body cells.



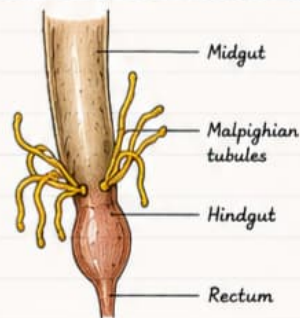
6. CIRCULATORY SYSTEM

- Open type circulatory system.
- Heart is a long, tube-like structure present dorsally in abdomen.
- Blood (haemolymph) is not red; it is colourless.
- Haemolymph bathes the organs directly.



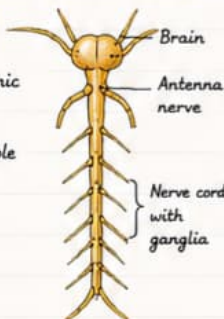
7. EXCRETORY SYSTEM

- Excretion is done by Malpighian tubules.
- They are numerous, yellow coloured filamentous structures.
- They open at the junction of midgut and hindgut.
- Main waste product - Uric acid.



8. NERVOUS SYSTEM

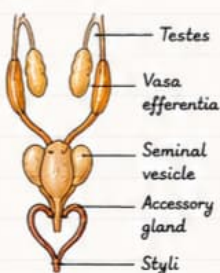
- Cockroach has a ganglionic nervous system.
- It consists of brain, double ventral nerve cord and segmental ganglia.
- From the brain arise antennae nerves.



9. REPRODUCTIVE SYSTEM

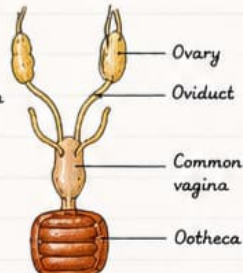
MALE

- Testes - 6 to 8
- Vasa efferentia open into seminal vesicle.
- Accessory glands present.
- A pair of styli present.



FEMALE

- Ovaries - 2
- Oviducts open into common vagina.
- Ootheca (egg case) is formed.



Key Takeaway



Cockroach shows all basic characters of arthropods. Its organ systems are well developed and efficient.

Why Cockroach is a Good Study Organism?

- ✓ Easy to maintain in laboratory.
- ✓ Short life cycle and high fecundity.
- ✓ Shows all major organ systems clearly.
- ✓ Helpful in understanding insect anatomy and physiology.

